

MEDSIM 3D

Medical Simulation Training Platform

Medical Simulation Training Platform

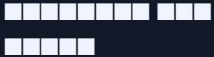
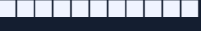
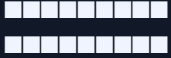
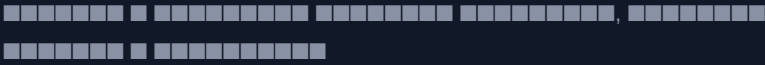
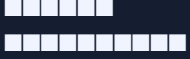
3D-Modeling

1.0 (2026)
2026
Web SaaS —
Next.js 14 / Three.js / Node.js / PostgreSQL
GLM-5.1 / DeepSeek V4 / Kimi K2.6
- , ,
READY FOR DEVELOPMENT

- React / Next.js
- Three.js 3D
- AI
- UI
- Admin Panel
- REST API



Rank	Project Name	Score
1	OpenAI GPT-4	3
2	Google Gemini Pro	4
3	Meta LLaMA 3	5
4	Claude 3.5 Sonnet	6
4.1	Gemini 1.5 Flash	6
4.2	3D Gemini Pro Vision	7
4.3	Gemini 1.5 Pro	8
4.4	Gemini 1.0 Ultra (Gemini 1.0 Pro)	9
4.5	AI-Gemini Pro (LLM)	10
4.6	Gemini 1.0 Nano	11
5	OpenAI GPT-4o	12
6	API OpenAI GPT-4o	13
7	OpenAI GPT-4o mini	14
8	OpenAI GPT-4o (Multimodal)	16
9	OpenAI GPT-4o (Text-to-Speech)	17
10	OpenAI GPT-4o (Code Interpreter) LLM (OpenCode)	18

		
		
	3D 	
	AI- 	LLM 
		
		

	HEX	
--bg-primary	#0A0F1E	
--bg-card	#111827	
--accent-cyan	#00E5FF	CTA
--accent-red	#FF3B5C	
--accent-orange	#FF7043	
--accent-yellow	#FFB547	
--accent-green	#00D68F	
--text-primary	#F0F4FF	
--text-secondary	#8892A4	
--border	#1E2D4A	

```
LOGO MedSim 3D [ ] [ ] [ ]... [Avatar] ← Topbar  
[ ]  
[ ]  
[ ] 3D - [ ] [ ]  
[ ] (Three.js canvas) [ ]  
[ ] • [ ] [ ] [ ]  
[ ] • [ ] [ ] [ ] : [ ]  
[ ] • [ ] 3D [ ] [ ] [_____] [ ]  
[ ] • t° 38.5°C [ ] [ ] [ ] [ ]  
[ ] [AI [ ]]  
[ ] : [ ] [✓]  
[ ] 52 [ ] [ ]  
[ ] 20 [ ] ← [ ] [ ] : 7/10 [ ]  
[ ] [ ] [ ] 70% [ ]
```

```
[██████████ - Next.js 14 + React 18]
██ HTTPS / WebSocket
▼
[API GATeWAY - Node.js / Express]
██ ██
▼ ▼
[PostgreSQL DB] [LLM Service]
[Redis Cache] GLM-5.1 / DeepSeek / Kimi
██
▼
[File Storage - S3 / MinIO]
(3D ██████████.glb, ████████████████████, PDF-██████████)
```

[illegible]

[illegible]

■■■■■■■■■■ ■■■■■■■■■■: JWT Access Token (15 ■■■) + Refresh Token (7 ■■■■, httpOnly cookie).

OAuth 2.0 (Google, GitHub) 2FA TOTP (Google Authenticator) —

Rigged-

:

(

,)

	()		
	,	#FF3B5C	
/	,	#FF3B5C	
	,	#FF7043	
	,	#FF3B5C	
()	,	#B040FF	+
()	,	#FFB547	

```
[ ]

```

▼

```
[ ] [3D: ]
• 
• 
• 

```

▼

```
[ ]
•  + autocomplete -11
•  ( )
•  ( )

```

▼

```
[AI (LLM) ]
• 
•  ( , , )

```

▼

```
[ ]
•  + 
• 
• 
•  +  = 

```

#	Input	Output
0		
1		
0		
2		
0		
3		
0		
4		
0		
5		
0		
6		

07		, , , ,
08		, , , ,
09		, ,
10		, , COVID-19, ,
11		, , , ,
12		, , ,
13		, , ,
14		, , ,
15		, ,
16		, , ,
17		, , ,
18		, , ,
19		, , ,
20		, ,
21		, , ,
22		, , ,
23		, , ,
24		, ,
25		, , , ,

4.5 AI-辅助诊断 (LLM)

本系统集成了 LLM 模型进行辅助诊断。支持的模型包括：GLM-5.1 — 讯飞星火, DeepSeek V4 Flash — 深度求索, Kimi K2.6 — 月之暗面。

LLM	模型	数据格式 (Prompt)
主界面 医生输入 患者信息	GLM-5.1 / DeepSeek	{case_data}, {student_answer}, {correct_diagnosis}
医生工作台 医生输入	GLM-5.1	{diagnosis_result}, {student_level}, {specialty}
主界面 患者输入 医生输入	Kimi K2.6	{case_context}, {student_question}
医生工作台 医生输入 (Admin)	Kimi K2.6	{specialty}, {difficulty}, {symptoms_list}
医生工作台 医生输入 患者信息	DeepSeek V4	{symptoms}, {vitals}, {anamnesis}
医生工作台 医生输入 患者信息	DeepSeek V4 Flash	{symptom_name}, {body_zone}, {severity}

4.6 系统功能模块

- 01. Dashboard: 系统概览 (用户统计, 课程统计, 考试统计, 系统健康)
- 02. 用户管理: 教师 / 学生 / 管理员
- 03. 课程管理: 新增课程, 编辑课程, 删除课程, 3D-模型, 课程资源, 课程评价/反馈
- 04. AI-辅助诊断: 教师 '输入患者信息' 模型 LLM → 输出诊断结果 → 教师
- 05. 考试管理: 新增考试, 编辑考试, 删除考试, 考试结果, 考试分析
- 06. 3D-模型: 上传模型 mesh-文件, 模型管理
- 07. LLM: API 配置, 模型选择, 模型测试, 模型训练
- 08. 系统设置: 系统配置, 模型配置, 日志管理

Frontend	Next.js	14.x	App Router, SSR, Server Components
UI	React	18.x	Hooks, Suspense, concurrent features
	TailwindCSS + CSS Variables	3.4+	,
3D	Three.js + React Three Fiber	r158 / 8.x	WebGL, GLTF,
3D helpers	Drei (pmndrs)	9.x	OrbitControls, HTML overlays, glow
State management	Zustand	4.x	,
	React Hook Form + Zod	7.x / 3.x	,
HTTP	Axios / TanStack Query	5.x / 5.x	, refetch, mutations
Backend	Node.js + Fastify	20 LTS / 4.x	, TypeScript-first
ORM	Prisma	5.x	,
	PostgreSQL	16.x	ACID, JSON-,
	Redis	7.x	, rate limiting, pub/sub
	JWT + bcrypt	—	Stateless,
	TypeScript	5.x	
	Vitest + Playwright	1.x / 1.x	Unit + E2E
CI/CD	GitHub Actions	—	,
	Docker + Compose	25.x	
	Nginx	1.25	SSL, ,
	MinIO (self-hosted S3)	—	3D ,

METHOD	ENDPOINT	
POST	/api/auth/register	
POST	/api/auth/login	, JWT
POST	/api/auth/refresh	access token
GET	/api/specialties	
GET	/api/specialties/:id/cases	
GET	/api/cases/:id	(, ,)
POST	/api/sessions	
POST	/api/sessions/:id/answer	→ AI
GET	/api/sessions/:id/feedback	
GET	/api/users/:id/stats	
GET	/api/users/:id/history	
POST	/api/admin/specialties	(Admin)
POST	/api/admin/cases	(Admin)
POST	/api/admin/cases/generate	AI- (Admin)
PUT	/api/admin/cases/:id	(Admin)
DELETE	/api/admin/cases/:id	(Admin)
GET	/api/admin/dashboard	(Admin)

```
users → id, email, password_hash, role, name, avatar, created_at
specialties → id, name, slug, icon, description, is_active, order
cases → id, specialty_id, title, difficulty(1-5), anamnesis,
vitals(JSON), symptoms(JSON), correct_diagnosis, explanation,
body_zones(JSON: [{zone_id, color, intensity}]), is_published
body_zones → id, mesh_name, label_ru, label_en, x, y, z (hint position)
sessions → id, user_id, case_id, started_at, finished_at, student_answer,
score, feedback(JSON), time_spent_sec
llm_logs → id, session_id, model, prompt_tokens, completion_tokens,
latency_ms, created_at
achievements → id, user_id, type, case_id, earned_at
```


UI

→ Responsive design: (3D → , → tabs)

→ Error Boundary ■ fallback UI

→ Dark/Light toggle (optional,          dark)

→ Accessibility: ARIA labels, keyboard navigation, contrast check

[illegible]

→ Vitest: unit LLM Router, Prisma ,

→ Playwright E2E: login flow,  , admin  

→ Load testing: k6 /sessions/:id/answer endpoint

→ Docker production build + multi-stage Dockerfile

→ Nginx config: SSL, gzip, static files, rate limiting

→ Seed : 50 5

```

    LLM: playwright E2E flow: →
    → → → . Page
    Object Model.

```


[illegible]

```
medsim/
├── apps/
│   ├── web/                                     # Next.js 14 frontend
│   │   ├── app/
│   │   │   ├── (auth)/login/                  # 
│   │   │   ├── (auth)/register/               # 
│   │   │   ├── (main)/specialties/           # 
│   │   │   ├── (main)/cases/[id]/            # (3D + )
│   │   │   ├── (main)/profile/               # 
│   │   │   ├── admin/                        # -
│   │   ├── components/
│   │   │   ├── robot/                        # Three.js 
│   │   │   │   ├── HumanRobot.tsx
│   │   │   │   ├── BodyZoneHighlight.tsx
│   │   │   │   ├── useBodyZones.ts
│   │   │   ├── case/
│   │   │   │   ├── CaseLayout.tsx
│   │   │   │   ├── SymptomsList.tsx
│   │   │   │   ├── DiagnosisForm.tsx
│   │   │   │   ├── FeedbackPanel.tsx
│   │   │   ├── admin/
│   │   │   │   ├── CaseWizard.tsx
│   │   │   │   ├── ZonePicker.tsx
│   │   │   ├── ui/                          # 
│   │   ├── lib/
│   │   │   ├── api.ts                        # Axios instance
│   │   │   ├── stores/                      # Zustand stores
│   │   ├── public/models/                   # .glb 3D 
│   │   └── 
│   └── api/                                  # Fastify backend
│       ├── src/
│       │   ├── routes/
│       │   │   ├── auth.ts
│       │   │   ├── specialties.ts
│       │   │   ├── cases.ts
│       │   │   ├── sessions.ts
│       │   │   ├── admin/
│       │   ├── services/
│       │   │   ├── llm/
│       │   │   │   ├── router.ts             # LLM 
│       │   │   │   ├── glm.ts
│       │   │   │   ├── deepseek.ts
│       │   │   │   ├── kimi.ts
│       │   │   ├── diagnosis/
│       │   │   ├── middleware/
│       │   │   │   ├── auth.ts
│       │   │   │   ├── roleGuard.ts
│       │   ├── prisma/
│       │   │   ├── schema.prisma
│       └── Dockerfile
├── packages/
│   ├── shared/                                # TypeScript
│   │   ├── types/
│   │   │   ├── case.ts
│   │   │   ├── user.ts
│   │   │   └── llm.ts
├── docker-compose.yml
├── docker-compose.prod.yml
└── .github/workflows/ci.yml
```

■■■ README.md

■■■■■■■■■■ ■■■■ ■■■■■■■■■■ ■■■■ ■■■■■■■■■■ ■ OpenCode ■ ■■■■■■■■■■ GLM / DeepSeek / Kimi.
■■■■■■■■■■ ■ ■■■■■■■■■■: '■■■■■■■■■■■ TypeScript. ■■■■■■■■■■: Next.js 14 + Fastify + Prisma + Three.js.'

[illegible]

```

■■■■■■■■■■ ■■■■■■■■■■ ■■■■■■■■■■ ■■■■■■■■■■ ■■■■ ■■■■■■■■■■ MedSim 3D. ■■■■■■■■■■: Next.js 14 (App
Router), Fastify, Prisma + PostgreSQL, Redis, TailwindCSS, TypeScript ■■■■■■■■■■. ■■■■■■■■■■
docker-compose.yml ■■■■■■■■■■: web, api, postgres, redis, minio. ■■■■■■■■■■ Prisma schema ■■■■■■■■■■
■■■■■■■■■■: users (id, email, password_hash, role enum, name), specialties, cases (■■■■■■■■■■
symptoms JSONB, body_zones JSONB, vitals JSONB), sessions, llm_logs.

```

#2 — 3D

```

████████ React Three Fiber ██████████ HumanRobot.tsx. ██████████ GLTF ██████████ █
/models/human.glb ██████████ useGLTF. ██████████ ███ useBodyZones(zones: Zone[]) ██████████
████████████████████ {meshName: string, color: string, intensity: number} █ ██████████ █
████████████████████ mesh ██████████ █ emissive + pulse ██████████ █████ useFrame. ██████████
OrbitControls, ██████████ ██████████ ██████████, Html tooltip ███ hover ███ ██████████.

```

■■■■■ #3 — LLM Router

```

##### LLM Router #####  ## TypeScript ### Fastify. #####  ## OpenAI-compatible
#####: GLM-5.1 (api.zhipuai.cn), DeepSeek V4 Flash (api.deepseek.com), Kimi K2.6
(api.moonshot.cn). #####  #####  ##  ##  #####: EVALUATE_DIAGNOSIS → GLM-5.1,
GENERATE_FEEDBACK → GLM-5.1, PATIENT_CHAT → DeepSeek, GENERATE_CASE → Kimi, DIFFERENTIAL →
DeepSeek. Fallback ##  #####  #####  ##  #####. #####  # ##### llm_logs.

```

■■■■■ #4 — ■■■■■■■■ ■■■■■

```

##### /cases/[id]/page.tsx ■ Next.js 14. Layout: #####. #####
(25%): SymptomsList ■ #####, AnamnesisCard ■ #####
##### , VitalsCard ■ #####/#####/#####. ##### (50%): 3D
HumanRobot ■ ##### case.body_zones. ##### (25%): DiagnosisForm ■
autocomplete #-11, ##### '#####', #####-## #####, ##### Submit.
TailwindCSS dark theme.
```

■■■■■ #5 — ■■■■■ ■■■■■ (Admin)

```
##### multi-step wizard ##### CaseWizard.tsx ### admin #####. ### 1: #####
##### (title, specialty_id, difficulty 1-5, description). ### 2: #####
(##### , vitals JSON #####, #####/#####
#####). ### 3: ##### 3D ##### (##### 3D #####, ##### → #####
#####/#####). ### 4: ##### '#####
##### AI' ### 1 - ##### POST /admin/cases/generate.
```

Page 10 of 10

~15

API endpoints: ~35+ REST routes

